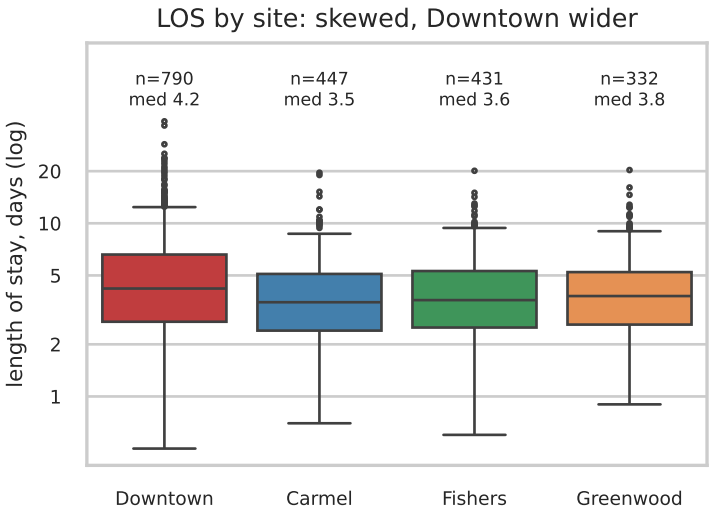
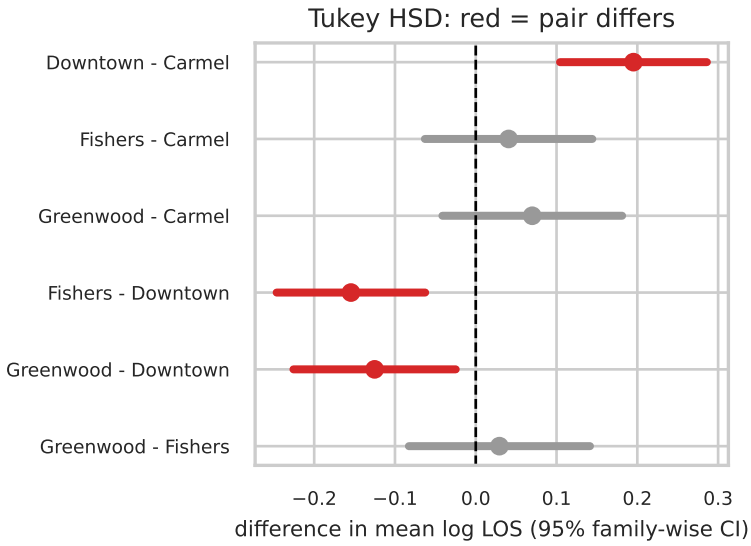
Day 9 Cheat Sheet: Comparing Many Groups

ANOVA, Kruskal-Wallis, post-hoc tests and the multiple-testing trap

TERM	WHAT IT MEANS	CODE
One-way ANOVA	are k group means all equal? (one F test)	stats.f_oneway(*groups)
F statistic	variance BETWEEN groups / variance WITHIN	big F = groups differ more than noise
Eta squared	share of variance explained by the group	SS_between / SS_total
Levene test	are group variances equal?	stats.levene(*groups)
Welch ANOVA	ANOVA without the equal-variance assumption	anova_oneway(y, g, use_var="unequal")
Kruskal-Wallis	rank-based ANOVA for skewed data / outliers	stats.kruskal(*groups)
Tukey HSD	which pairs differ, family-wise error at 5%	pairwise_tukeyhsd(y, g)
Holm / Bonferroni	adjust p-values for many tests (FWER)	multipletests(p, method="holm")
Benjamini-Hochberg	control the false discovery rate (FDR)	multipletests(p, method="fdr_bh")
Two-way ANOVA	two factors + their interaction	anova_lm(ols("y ~ A * B"), typ=2)





THIS DATA (simulated): 2,000 inpatient stays at 4 sites, mean LOS 4.61 days, median 3.8 days
One-way ANOVA F = 20.90, p = 2e-13, but eta squared = 0.030: site explains only 3% of LOS variation
Assumptions fail: Levene p = 7e-15 (Downtown SD 4.09 vs ~2.4), residual skew 2.87 -> Welch F = 17.60, p = 4e-11
Kruskal-Wallis H = 33.13, p = 3e-07; ANOVA on log LOS F = 12.55, p = 4e-08. Every version agrees
Tukey + Holm: Downtown differs from all 3 suburban sites; Carmel, Fishers, Greenwood do not differ from each other

HABITS THAT KEEP YOU HONEST
- ANOVA only says "not all equal"; run a post-hoc test to say which pairs differ
- Never run all pairwise t-tests at 0.05 and report the hits; adjust the p-values
- Check spread and skew first (Levene, histogram); Welch or Kruskal-Wallis when they fail
- Report the effect size (eta squared) and the gap in days, not just a tiny p-value
- Decide how many comparisons you'll make BEFORE you look at the data

```
groups = [g["los_days"] for _, g in df.groupby("site")]
F, p = stats.f_oneway(*groups);  H, p_kw = stats.kruskal(*groups)
tk = pairwise_tukeyhsd(np.log(df["los_days"]), df["site"]);  print(tk.summary())
reject, p_adj, _, _ = multipletests(p_values, method="holm")
```

Infographic: One Test, Then the Right Pairs

Simulated: 2,000 inpatient stays, length of stay (days) at 4 sites

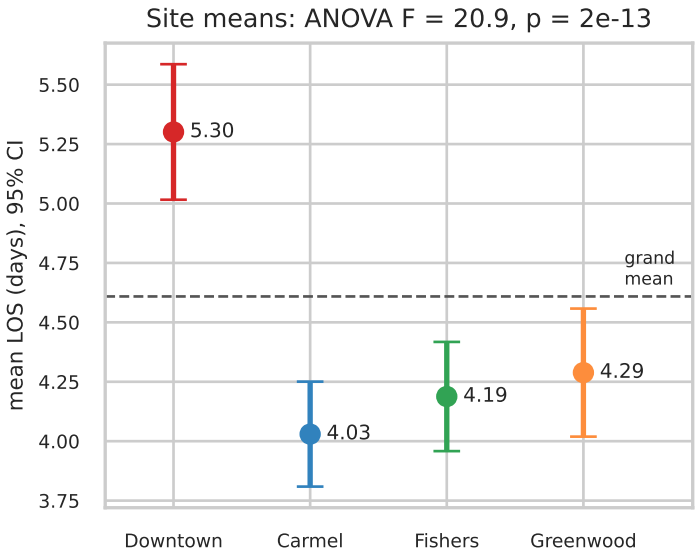
Describe	Check	Omnibus	Size	Post-hoc	Adjust
n, mean, median, SD	Levene, skew	ANOVA / Welch / Kruskal	eta squared, gap in days	Tukey HSD or Holm	many metrics? FDR (BH)

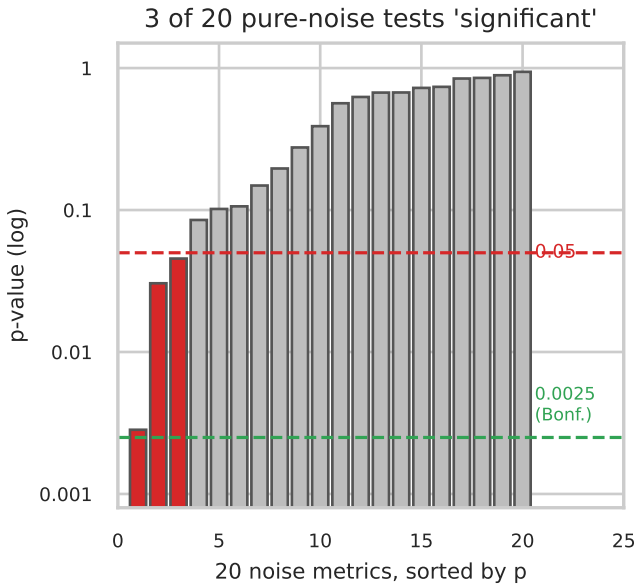
ANOVA
F 20.9
p = 2e-13

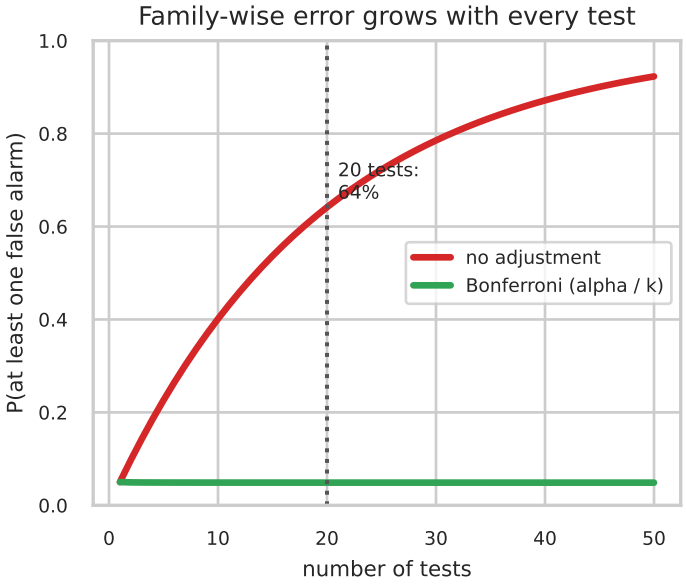
ETA SQUARED
0.030
small effect

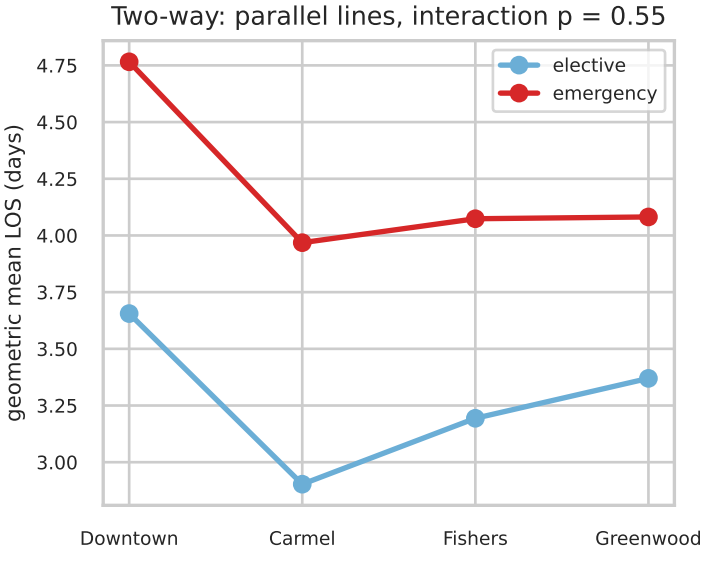
DOWNTOWN GAP
+1.1 days
mean LOS

NOISE HITS
3 of 20
0 after adjust









Read it like a manager: Downtown stays run about 1 day longer on average, the three suburban sites look alike, and emergency admissions are longer at every site by about the same proportion (no interaction). Simulated, observational data.

Hands-On Worksheet: ds-genai-90-days / day09

60-90 minutes | notebook: day09/los_anova.ipynb

[ ] mkdir day09, copy day07/patients.csv in, conda activate ds90

[ ] Simulate stays.csv with seed 9 (snippet 1); mean LOS should be 4.61

[ ] groupby site: count, mean, median, SD. Which site looks different?

[ ] Boxplot LOS by site on a log axis; note skew and spread

[ ] stats.f_oneway: F, p. Compute eta squared by hand (0.030)

[ ] Levene + residual skew: which ANOVA assumption fails?

[ ] Welch ANOVA (anova_oneway, use_var='unequal'): same story?

[ ] stats.kruskal and ANOVA on log LOS; compare geometric means

[ ] pairwise_tukeyhsd on log LOS: which pairs reject?

[ ] 6 pairwise Mann-Whitney tests, then multipletests(method='holm')

[ ] 20 noise metrics: count raw hits, then Bonferroni, Holm, BH

[ ] Two-way ANOVA site * admit_type; read the interaction p-value

[ ] Findings cell; git add day09 && git commit && git push

[ ] Stretch: Snowflake LOS summary by site with MEDIAN and PERCENTILE_CONT

```
import numpy as np, pandas as pd
from scipy import stats
from statsmodels.stats.multicomp import pairwise_tukeyhsd
from statsmodels.stats.multitest import multipletests
df = pd.read_csv("stays.csv")
groups = [g["los_days"] for _, g in df.groupby("site")]
print(stats.f_oneway(*groups))                 # expect F ~20.90
print(stats.kruskal(*groups))                  # expect H ~33.13
print(pairwise_tukeyhsd(np.log(df["los_days"]), df["site"]).summary())
```

Mean LOS: _____ F: _____ p: _____ eta sq: _____ Levene p: _____ Welch F: _____

Kruskal H: _____ Tukey pairs that differ: _____ Noise hits raw: _____ after Holm: _____

Day 9 Self-Check Quiz

1. Why not just run six t-tests to compare four sites?

   _____

2. ANOVA gives F = 20.9, p < 0.001. What can (and can't) you conclude?

   _____

3. Eta squared is 0.030. How do you explain that to a manager?

   _____

4. Levene p is tiny and residual skew is 2.87. What do you do?

   _____

5. What does Kruskal-Wallis compare, and when do you reach for it?

   _____

6. Tukey HSD rejects 3 of 6 pairs. Summarize the pattern in one sentence.

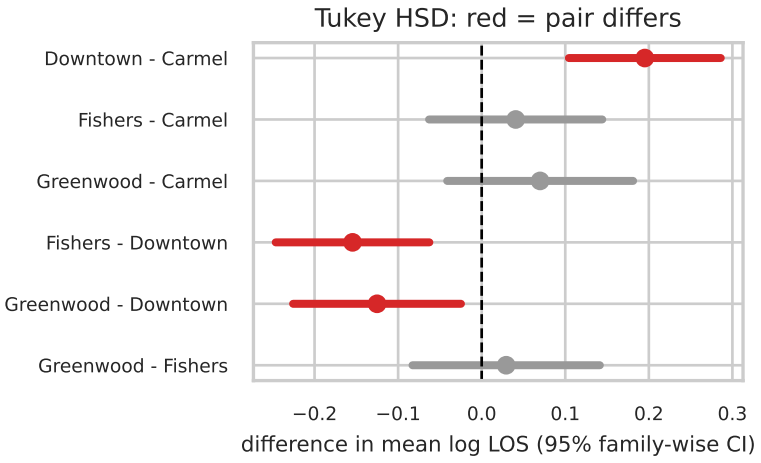
   _____

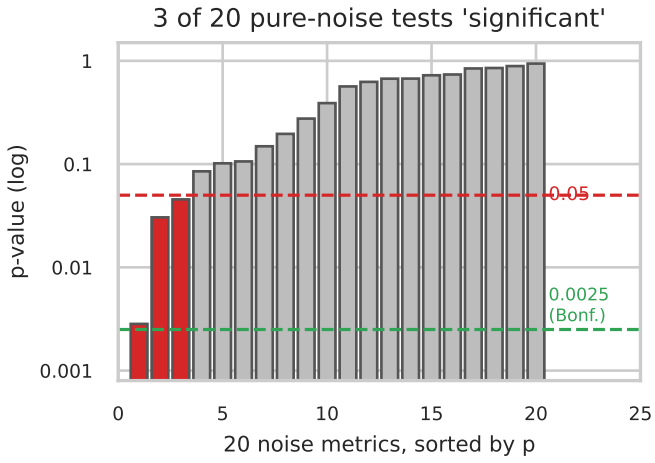
7. 3 of 20 noise metrics have p < 0.05. Why, and what fixes it?

   _____

8. Holm/Bonferroni vs. Benjamini-Hochberg: when would you choose each?

   _____





Answers:
1 each test has a 5% false-alarm rate; 6 tests push the chance of at least one false alarm to about 26%. One ANOVA asks one question
2 at least one site mean differs from the others; it does NOT say which one, how big, or why (site is not randomized)
3 site explains about 3% of the variation in LOS; real, but most of the variation is patient-level (admit type, chronic conditions)
4 use Welch ANOVA (F 17.6) or Kruskal-Wallis, or analyze log(LOS); here all agree, so the finding is robust
5 ranks across the groups (whether one group tends to have larger values); for skewed data, outliers or ordinal scores like pain 0-10
6 Downtown stays are longer than at each suburban site, and the three suburban sites are indistinguishable
7 at alpha 0.05 you expect 1 false alarm per 20 tests by chance; adjust the p-values (Holm, Bonferroni, or BH) and all 3 disappear
8 Holm/Bonferroni when any false positive is costly (a few key comparisons); BH when screening many metrics and some false hits are OK